

Positivity Beyond the First Thue Morse Feedback Boundary

Research note prepared for Vladimir Reshetnikov with OpenAI

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Abstract

We extend the positive diagonal asymptotic for integer Thue–Morse pressure responses to every fixed offset beyond the first nonlinear feedback boundary. A capped inverse-orbit expansion and a marked-pair estimate separate the positive nearest orbits from an exponentially smaller remote tail. The restored full-degree insertion and eigenvalue feedback have only polynomial size at a fixed offset. Quantitative bounds also give an explicit growing region: both the response and pressure coefficients at offset s beyond the boundary are positive under either $m \geq 800(s+1)^2$ or the stronger large- m condition $m \geq 2048$ and $200(s+1)\log(2m) \leq m$. The proposed full pressure range strictly below degree $6m$ remains open.

Status. This is unrefereed ordinary mathematics with exact arithmetic checks, not a Lean formalization. The earlier positivity-triangle and boundary reports remain separate. No priority claim or formula for the first later negative pressure coefficient is made.

1 Results and normalization

Set $d = 2m$, $m \geq 2$, and use the sum-over-preimages operator

$$(V_a f)(x) = \sum_{2y=x \bmod 1} (\cos \pi y + a \sin \pi y)^d f(y).$$

Its local analytic eigenfunction is normalized by $h_a(0) = 1$. Put

$$R_m = \sum_{w \in \mathbb{Z}+1/2} (\pi w)^{-2m}, \quad H_{m,r} = [a^{2r}]h_a(1/2), \quad E_{m,r} = [t^{2m+2r}]p_m(t/\pi).$$

The pressure and eigenvalue obey

$$\lambda_m(a) = 1 + a^d h_a(1/2), \quad p_m(t/\pi) = d \log \cos t + \log \lambda_m(\tan t). \quad (1)$$

Previous reports prove $H_{m,r}, E_{m,r} > 0$ for $1 \leq r \leq m$ [2, 3]. Define

$$b_j = \tan \frac{\pi}{2^{j+1}}, \quad B(a) = \prod_{j \geq 1} (1 + ab_j), \quad \frac{a_* B'(a_*)}{B(a_*)} = 1,$$

$$C = \frac{B(a_*)}{a_*}, \quad p_j = \frac{a_* b_j}{1 + a_* b_j}, \quad \sigma^2 = \sum_{j \geq 1} p_j (1 - p_j).$$

The product converges locally uniformly, and its mean function increases strictly from zero to infinity. The constants are approximately $a_* = 0.8229185691$, $C = 4.046860022$, and $\sigma^2 = 0.7041449521$; these decimals are not used in the proof.

Theorem 1.1. *For every fixed integer $s \geq 0$,*

$$H_{m,m+s} \sim \frac{2(2C/\pi)^{2m} a_*^{-2s}}{\sqrt{4\pi m \sigma^2}} \quad (m \rightarrow \infty). \quad (2)$$

Consequently $H_{m,m+s}$ and $E_{m,m+s}$ are positive for all sufficiently large m . Explicitly, for all integers $s \geq 0$,

$$m \geq 800(s+1)^2 \implies H_{m,m+s} > 0, \quad E_{m,m+s} > 0. \quad (3)$$

The same conclusion holds under the alternative condition

$$m \geq 2048, \quad 200(s+1) \log(2m) \leq m. \quad (4)$$

Thus the pressure range extends beyond degree $4m$ by an unbounded number of even degrees. The logarithmic condition allows offsets of order $m/\log m$. Both cutoffs are deliberately loose. The statement does not prove positivity throughout $2m < \deg < 6m$. The eventual scope for responses is necessary: $H_{2,3} = -488/27$, whereas its degree-10 pressure counterpart is positive [3].

2 An exact two-cluster expansion

Work in the invariant Fourier space with indices $1-m, \dots, m-1$. Let $A = V_0$, $h_0 = h_{a=0}$, $Pf = f(0)h_0$, $Q = I - P$, and $\text{ev } f = f(1/2)$. The atomic basis

$$F_{j,m}(x) = \sin^{2m}(\pi x) \sum_{n \in \mathbb{Z}} (\pi(x+n))^{-(2m-j)}, \quad 0 \leq j \leq 2m-2,$$

satisfies $AF_{j,m} = 2^{-j}F_{j,m}$ and $h_0 = F_{0,m}$. Separating even and odd terms in the periodization proves these identities and the simple-eigenvalue setup [1, 2].

On Q set

$$B_a = QV_aQ, \quad q_a = QV_a h_0. \\ \mathcal{F}(a) = R_m + \text{ev}(I - B_a)^{-1}q_a, \quad \mathcal{G}(a) = \text{ev}(I - B_a)^{-2}q_a.$$

Both scalar series are even, $\mathcal{F}(0) = R_m$, and $\mathcal{G}(0) = 0$. They describe the response with eigenvalue fixed at one and its extra resolvent response.

Lemma 2.1. *As analytic germs,*

$$\log \lambda_m(a) = a^d \mathcal{F}(a) - a^{2d} \left(\mathcal{F}(a)\mathcal{G}(a) + \frac{\mathcal{F}(a)^2}{2} \right) + O(a^{3d}). \quad (5)$$

Proof. Write $h_a = h_0 + u_a$, $u_a \in Q$, and $\delta = \lambda_m(a) - 1$. Applying Q gives $(I - B_a + \delta)u_a = q_a$. Thus

$$\delta = a^d \{R_m + \text{ev}(I - B_a + \delta)^{-1}q_a\} = a^d (\mathcal{F} - \delta\mathcal{G} + O(\delta^2)).$$

Substitute once and expand $\log(1 + \delta)$. □

Substitution $a = \tan t$ preserves the remainder order, so only two feedback clusters matter below degree $6m$. Raw frozen positivity is false:

$$\mathcal{F}_2(a) = \frac{1}{3} + 8a^2 + \frac{536}{27}a^4 - \frac{296}{27}a^6 + O(a^8).$$

Its tangent composition nevertheless gives positive pressure coefficients of degrees 6, 8, 10, respectively:

$$376/45, \quad 6836/189, \quad 105448/2025.$$

3 Capped orbits and marked pairs

Write $V_a = \sum_{j=0}^d a^j V_j$ and remove the full-degree insertion:

$$V_{<}(a) = \sum_{j=0}^{d-1} a^j V_j.$$

Since $(V_j f)(0) = 0$ for $1 \leq j < d$, it has a local analytic fixed vector $f_{<}(a)$ with $f_{<}(a)(0) = 1$. Put $Z(a) = \text{ev } f_{<}(a)$. Let $g(x) = \sin \pi x$ on $[0, 1]$, extended periodically and nonnegative. The elementary branch estimates are

$$Ag \leq \frac{1}{2}g, \quad |V_j f| \leq \frac{1}{2} \binom{d}{j} \|f\|_{\infty} g \quad (1 \leq j < d), \quad \|h_0\|_{\infty} \leq 1. \quad (6)$$

For the middle bound, set $c = \cos(\pi x/2)$ and $u = \sin(\pi x/2)$, factor cu from $c^{d-j}u^j + u^{d-j}c^j$, and use weighted AM–GM to bound the remaining sum by $c^{d-2} + u^{d-2} \leq 1$. The first bound is similar. The last follows from the positive polynomial expansion of $h_0(x + 1/2)$ in $\sin^2 \pi x$ [3].

For a summable nonnegative sequence x define

$$\mathcal{C}_{d,q}(x) = [z^q] \prod_i ((1 + zx_i)^d - (zx_i)^d).$$

Finite-iterate periodization and coefficientwise convergence give

$$[a^q]Z(a) = \sum_{w \in \mathbb{Z} + 1/2} (\pi w)^{-d} [a^q] \prod_{j \geq 1} ((1 + at_j(w))^d - (at_j(w))^d), \quad (7)$$

where $t_j(w) = \tan(\pi w/2^j)$. For each fixed d, q this series is absolutely convergent. Expanding into finitely many compositions of q , every insertion degree is below d , and the zero-degree gaps form geometric sums by (6). The bounded initial vector h_0 enters the g -norm space at its first positive-degree insertion; it is never treated as a vector vanishing at zero.

Lemma 3.1. *Let $q = d + 2s$, $k = 2s + 1$, and $d \geq 2s + 2$. For every $a > 0$,*

$$\mathcal{C}_{d,q}(x) \leq \frac{d^{2k}}{k!} a^{2k-q} e_2(x)^k B_x(a)^d, \quad B_x(a) = \prod_i (1 + ax_i). \quad (8)$$

Proof. First use finitely many coordinates. A retained monomial has q labeled units with color multiplicities $\alpha_i \leq d - 1$. Its maximum number of disjoint pairs with distinct colors is

$$\min(\lfloor q/2 \rfloor, q - \max_i \alpha_i) \geq 2s + 1 = k.$$

This formula follows by pairing between the two largest remaining classes; if one class exceeds half the units, pair every unit outside it with a unit inside it.

Introduce independent variables y_i and

$$\mathcal{D}_x = \sum_{i < j} x_i x_j \frac{\partial^2}{\partial y_i \partial y_j}.$$

In the nonnegative polynomial

$$\frac{z^{2k}}{k!} \mathcal{D}_x^k \prod_i (1 + y_i)^d \Big|_{y_i = zx_i},$$

a monomial's coefficient is its original coefficient multiplied by the number of unordered matchings of k cross-color pairs among its labeled units. The derivatives choose distinct units; the $k!$ orders of the pairs cancel the denominator. This integer multiplier is at least one for every retained monomial.

At $z = a$, each derivative costs at most $dx_i/(1 + ax_i)$, with repeated falling factorials bounded by powers of d . Summing the pairs and dropping denominators bounds the value by $d^{2k}a^{2k}e_2(x)^k B_x(a)^d/k!$. Divide by a^q to bound its degree- q coefficient. Monotone convergence extends the result to summable sequences. $\square$

For positive odd n set $x_j(n) = |\tan(n\pi/2^{j+1})|/n$ and

$$S_s = \sum_{\substack{n \geq 3 \\ n \text{ odd}}} n^{2s} e_2(x(n))^{2s+1}.$$

Lemma 3.2. *For every $s \geq 0$, writing $z = s + 1$, we have*

$$S_s < \infty, \quad S_s \leq 2^{22z} z^{8z}. \quad (9)$$

Proof. Put $e = 2s + 2 = 2z$, $k = e - 1$. Algebraically,

$$n^{2s} e_2(x(n))^k = n^{-e} e_2(|\tan(n\pi/2^{j+1})| : j \geq 1)^k.$$

The numerator has degree $2k$, with each exponent at most $k = e - 1$. Its comparison with the capped product of exponent e costs at most $(2k)!$, which bounds the pair arrangements of the $2k$ units. Thus the auxiliary absolute insertion argument proves convergence.

For $s \geq 1$, sum all zero-degree gaps using (6). The resulting bound is

$$[u^{2k}] \frac{1}{1 - \sum_{j=1}^{e-1} \binom{e}{j} u^j} \leq \frac{3}{2} (4e)^{2k},$$

since at $u = 1/(4e)$ the inner sum is at most $\sum_{j \geq 1} 4^{-j} = 1/3$. Changing from half-integers to positive odd integers gives

$$S_s \leq \frac{3}{4} (\pi/2)^e (2k)! (4e)^{2k} \leq 2^{22z} z^{8z},$$

using $(2k)! \leq (4z)^{4z}$ and $(4e)^{2k} \leq (8z)^{4z}$. For $s = 0$, use $A_1 g \leq g/\sqrt{2}$; the direct two-insertion estimate gives $S_0 < 11$ [3], sufficient for (9). $\square$

4 The fixed-offset asymptotic

The strict tangent theorem in [3] says that $x(n)$ is weakly log-majorized by $(\theta, b_2, b_3, \dots)$ for odd $n \geq 3$, where $\theta = (1 + \sqrt{2})/3$. Hence

$$B_{x(n)}(a) \leq \eta(a) B(a), \quad \eta(a) = \frac{1 + \theta a}{1 + a} < 1. \quad (10)$$

The quantitative reason is that the sorted normalized tangent ratios cross one only once. Before the crossing their first ratio is at most θ and the rest are at most one. After it, the first r sorted terms are the original prefix, with product ratio $s_r^2 \prod_{j=2}^{r-1} s_j$, where $s_j = |\sin(n\pi/2^{j+1})|/(n \sin(\pi/2^{j+1})) \leq 1$ and $s_2 \leq \theta$. Convex symmetric exponential sums applied to logarithms give (10); geometric tails justify the infinite limit.

Fix s , with $q = d + 2s$ and $k = 2s + 1$. The entire remote contribution in (7) is, by (8)–(10), bounded by

$$2(2/\pi)^d \frac{d^{2k}}{k!} a_*^{2k-2s} C^d \eta(a_*)^d S_s. \quad (11)$$

The factor n^{2s} is precisely $n^q n^{-d}$ after rescaling the tangents. For fixed s , the remote sum is therefore exponentially smaller than the nearest scale, up to a polynomial in d .

The nearest pair contributes $2(2/\pi)^d \mathcal{C}_{d,q}(b)$. Because $q < 2d$, the full-degree monomials removed from $[u^{d+2s}]B(u)^d$ are exactly

$$\sum_j b_j^d [u^{2s}] \prod_{i \neq j} (1 + ub_i)^d = O_s(d^{2s}).$$

Indeed, $\sum b_j^d < 2$, $\sum b_j \leq 2$, and $[u^{2s}]B(u)^d \leq (2d)^{2s}/(2s)!$.

Let X have probability generating function $B(a_*u)/B(a_*)$, mean one, and variance σ^2 . It is a countable Bernoulli sum with all exponential moments and lattice span one. Fourier inversion gives

$$\mathbb{P}(X_1 + \dots + X_d = d + 2s) \sim (2\pi d\sigma^2)^{-1/2}.$$

The centered characteristic function is $1 - \sigma^2 v^2/2 + O(v^3)$ near zero, with a Gaussian modulus bound there and modulus strictly below one on the compact complement. Rescaling by $\sqrt{d}$ proves the estimate; the fixed shift contributes a factor tending to one. Thus

$$[u^{d+2s}]B(u)^d \sim C^d a_*^{-2s} / \sqrt{2\pi d\sigma^2}. \quad (12)$$

This establishes the asymptotic for $[a^{d+2s}]Z$.

To restore the full insertion, put $T_< = (I - QV_<Q)^{-1}$ on Q . The frozen vector satisfies $f_{\mathcal{F}} = f_< + a^d T_< QV_d f_{\mathcal{F}}$. For $2s < d$, the Schur expansion yields the exact identity

$$H_{m,m+s} = [a^{d+2s}]Z + [a^{2s}](\text{ev } T_< QV_d f_< - \mathcal{FG}). \quad (13)$$

Use $\|f\|_g = \sup |f|/g$ for vectors vanishing at zero. The resolvent $(I - A)^{-1}$ has g -norm at most two. Recursion gives order- j response and resolvent jets of size $O_j(d^j)$, for fixed $j < d$. The initial h_0 is handled in sup norm, not g -norm.

Also $\|QV_d f\|_{\infty} \leq 2\|f\|_{\infty}$. For a trigonometric polynomial of degree at most $m-1$, elementary Fourier coefficient bounds give $\|u'\|_{\infty} \leq 2\pi m(m-1)\|u\|_{\infty}$. Since $QV_d f$ vanishes at zero and $g(x) \geq 2 \text{dist}(x, \mathbb{Z})$,

$$\|QV_d f\|_g < 2d^2 \|f\|_{\infty}.$$

The restoration error in (13) is therefore $O_s(d^{2s+2})$ and the feedback error $O_s(d^{2s})$. They are negligible compared with $(2C/\pi)^d/\sqrt{d}$, proving (2).

5 An explicit growing strip

Assume $d \geq 1600z^2$, $z = s+1$, and $q = d+2s$. Choose a so that $aB'(a)/B(a) = q/d$. The earlier rational half-angle certificate gives $a > 4/5$ [3]. Moreover $q/d < 33/32$ and

$$B'(1)/B(1) > \frac{1}{2} + \frac{3}{11} + \frac{3}{19} + \frac{3}{35} + \frac{3}{67} > 33/32,$$

so $a < 1$. Thus $\eta(a) < 206/225$, since $\theta < 81/100$. Retaining $b_1 = 1$, $b_2 > 2/5$, and $b_3 > 1/6$ gives

$$\frac{B(a)}{a^{q/d}} \geq \frac{B(a)}{a} > \frac{1}{a} + \frac{47}{30} + \frac{19a}{30} + \frac{a^2}{15} > 3.$$

The a -tilted sum of d countable Bernoulli sums has integer mean q and variance at most q . The mean is a mode by the countable Darroch theorem [4, Theorem 2]. Chebyshev puts at least $3/4$ of its mass in an interval with at most $4\sqrt{q} + 1$ integers. Therefore, with $G = [u^q]B(u)^d$,

$$G \geq \frac{B(a)^d a^{-q}}{8\sqrt{q}}, \quad E_0 := 2(2/\pi)^d G > \frac{(21/11)^d}{4\sqrt{q}}.$$

Here E_0 is the unmodified nearest-orbit contribution, not a pressure coefficient; the last inequality uses $\pi < 22/7$.

Each of the following errors is less than $E_0/8$.

1. The remote relative error is bounded by

$$12\sqrt{d} d^{4z} 2^{22z} z^{8z} (206/225)^d.$$

Its logarithm divided by d is at most

$$\frac{\log 12 + \frac{1}{2} \log d}{d} + \frac{8 \log d + 22}{40\sqrt{d}} - \frac{19}{225} \leq \frac{93}{1600} - \frac{19}{225} < -\frac{1}{40}.$$

Use $z \leq \sqrt{d}/40$, $\log 2 < 1$, $\log 1600 < 8$, and $\log 12 < 3$. The positive functions are decreasing for $d \geq 1600$.

2. Deleting the nearest full-degree monomials has relative cost at most

$$24\sqrt{d} (2d)^{2z} 3^{-d}.$$

Its logarithm divided by d is at most $26/1600 - 1 < -1/2$, using $\log 24 < 4$ and $\log 3 > 1$.

3. An explicit low-order version of the norm recursion gives

$$\|f_n\|_g \leq (2d)^n \ (n \geq 1), \quad \|(T_{<})_n\|_g \leq 2(2d)^n, \quad \|\mathcal{G}_n^{\text{vector}}\|_g \leq 2n(2d)^n.$$

This follows by $\binom{d}{j} \leq d^j$ and summing compositions; $f_0 = h_0$ has sup norm at most one. The restoration and feedback error is at most $5d^2(2s+1)(2d)^{2s}$. Relative to E_0 it is at most

$$\frac{3}{2} d^3 (2d)^{2z} (11/21)^d.$$

Its logarithm divided by d is at most $43/1600 - 10/21 < -1/4$, using $\log(21/11) > 10/21$.

Thus

$$H_{m,m+s} > E_0/2 > (21/11)^d / (8\sqrt{q}) > 0. \quad (14)$$

The elementary monotonicity formulas and rational side inequalities are recorded in the verifier.

5.1 A wider logarithmic strip

The same proof works when $d \geq 4096$ and $400z \log d \leq d$. The saddle bounds remain valid since $q/d \leq 1 + 2/(400 \log d) < 33/32$. Now use $\log z \leq \log d$ in the remote error. Its logarithm divided by d is at most

$$\frac{\log 12 + \frac{1}{2} \log d}{d} + \frac{12z \log d}{d} + \frac{22z}{d} - \frac{19}{225} \leq \frac{7}{4096} + \frac{12}{400} + \frac{22}{3200} - \frac{19}{225} < -\frac{1}{40}.$$

Here $8 < \log 4096 < 9$ and $\log 12 < 5/2$. The nearest-deletion bound gives instead $17/8192 + 1/200 + 1/1600 - 1 < -1/2$. For restoration retain the original relative bound and use

$$20d^2(2s+1)\sqrt{q}(2d)^{2s}(11/21)^d \leq 60d^{7/2}(2d)^{2z}(11/21)^d.$$

Its logarithm divided by d is at most $73/8192 + 1/200 + 1/1600 - 10/21 < -1/4$. Thus (14) also holds under (4). The fixed- a_* asymptotic (2) remains a fixed- s assertion; no uniform extension of that particular formula is implied.

6 Pressure transfer and verification

Below degree $3d$, the true response gives

$$E_{m,m+s} = \sum_{j=0}^{m+s} H_{m,j}[t^{2(m+s-j)}](\tan t/t)^{d+2j} - \frac{1}{2}[t^{2s}](\tan t/t)^{2d} h_{\tan t}(1/2)^2 - \frac{m}{2m+s} R_{2m+s}. \quad (15)$$

For fixed s , all responses in the first line are eventually positive: use the earlier triangle for $j < m$ and (2) for the finitely many offsets $j = m, \dots, m+s$. The subtractions involve fixed-order jets and are polynomially bounded, while $H_{m,m+s}$ grows exponentially.

Under the explicit condition $d \geq 1600(s+1)^2$, every smaller offset obeys the same condition. Put $\rho = 1/(8d)$. The absolute low-order response series at $a = \tan \rho$ is bounded by $\sum_{n \geq 0} (2d \tan \rho)^n < 2$. Also $(\tan \rho/\rho)^{2d} < 2$, since $\tan \rho/\rho \leq (1-\rho^2)^{-1}$ and $(1-\rho^2)^{2d} \geq 1-2d\rho^2 > 1/2$. The quadratic subtraction is at most $4(8d)^{2s}$ and the cosine subtraction at most one. The positive term (14) exceeds their sum because

$$40\sqrt{q}(8d)^{2s}(11/21)^d \leq 60\sqrt{d}(8d)^{2z}(11/21)^d < 1.$$

Its logarithm divided by d is at most $31/1600 - 10/21 < 0$, using $\log 60 < 5$ and $\log 8 < 3$. Under the logarithmic condition, the same pressure bound has normalized logarithm at most $19/8192 + 1/200 + 3/1600 - 10/21 < 0$. This proves Theorem 1.1.

The bundle contains exact Fourier recursions checking all 195 pressure coefficients in $2m < \deg < 6m$ for $m = 2, \dots, 14$, and 35 independent checks of (13) for all $s < m$, $m = 2, \dots, 8$. It includes a finite-coordinate matching check, the rational strip verifier, and the earlier half-angle certificate locating the saddle. These calculations check identities and constants; the universal results follow from the analytic inequalities.

The full conjecture $E_{m,r} > 0$ for every $1 \leq r < 2m$ remains open. Finite first-negative-degree data and the heuristic rate near 6.662966658 do not enter the proof. Uniform sharp cutoffs or a formula for those first negative degrees require further control of the feedback clusters.

References

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